

DANMARKS TEKNISKE UNIVERSITET



MODELING CO₂ CONCENTRATIONS USING HIDDEN
MARKOV MODELS

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SPECIAL COURSE IN STATISTICAL MODELLING

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AI Declaration

AI completion has been on when writing code and no Agentic AI has been used in the written of the source code. Agentic AI has been used to write LaTeX formatting code and plots in the drivers directory. Agnetic AI has been used to write some of the tests in the tests directory. AI has been used for formatting LaTeX tables, figures and reformatting text in the report. All academic content, including the mathematical equations, h as been written by the author without the use of AI.

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1 Introduction

Hidden Markov Models (HMMs) are a powerful tool for modeling sequential data, where the underlying states of the system are not directly observable. They have been widely used in various fields such as speech recognition, natural language processing, and bioinformatics. (Ultralytics LLC, 2024).

The goal of this project is to inference if a person is in a room based on CO₂ data collected from a sensor in the room. To do this, we will use a Hidden Markov Model to model the relationship between the observed CO₂ levels and the hidden states of whether a person is in the room or not and if a window is open or not. When training the models, we will use **JAX** (Bradbury et al., 2018) and **EQUINOX** (Kidger and Garcia, 2021) for automatic differentiation and optimization, and we will use **Optax** (Hessel et al., 2020) as our optimization library.

WRITE HERE THE RESULTS YOU FIND LATER ON.

This report is written for the purpose of given a concise outline of the projects achivements, design choices, model results and interpretaiton. The report is written for future students to be able to easily carry on the the project and extend it in some way.

2 Software

This section will give an explanation of the classes implemented, the design choices made and the primary libraries used. A repository link can be found in 8.1 and the environment details and short source code overview can be found in 8.2. Class diagrams can be found in 8.4.

2.1 Class overview

The source code is organized into 6 main components: the `HMM` class, the `Emissions` class, the `Transitions` class, the `HMMParams` class, the `Inference` class, and the `Solvers` class. All classes except `HMM` are abstract interfaces. The `HMM` class acts as the main entry point, providing a simple interface for training and inference by delegating to the other components. The key design principle is that concrete implementations of `Emissions` and `Transitions` are injected at runtime, allowing different HMM variants to be used interchangeably. A detailed explanation of each class and the motivation behind the design choices can be found in 8.5, and class diagrams are provided in 8.4.

2.2 Design Principles

An Object-Oriented Programming (OOP) design was chosen for this project to promote modularity, extensibility, and separation of concerns. The Dependency Inversion Principle (DIP) was central to the design of all core components. They are defined as abstract interfaces, so different implementations can be swapped in without affecting the rest of the system. This modularity is also motivated by the mathematical structure of HMMs themselves. The forward algorithm, and HMM inference in general, depends only on two quantities. The emission density $p(y_t | x_t)$ and the transition probability $p(x_t | x_{t-1})$. It is entirely agnostic to how those quantities are computed. By encapsulating each behind an abstract interface, different HMM variants can be used interchangeably without modifying the inference or optimization code, so modularity follows naturally from the structure of the model. This also makes it straightforward to extend the project by adding new emission distributions, transition models, or inference algorithms without modifying existing code.

2.3 Primary Libraries

The primary libraries used in this project are `JAX` (Bradbury et al., 2018), `EQUINOX` (Kidger and Garcia, 2021), and `Optax` (Hessel et al., 2020).

`JAX` was chosen over `NumPy` for its automatic differentiation capabilities and performance optimizations. However, `JAX` is designed for functional programming and does not natively support classes or object-oriented design. Therefore, `EQUINOX` was chosen as it's a library built on top of `JAX` that provides a simple and flexible interface for defining classes as pytrees.¹, `JAX` can then automatically differentiate through the classes and the leaves of the pytree.

`Optax` was chosen as the optimization library for its seamless integration with `JAX` and `EQUINOX`, providing a wide range of optimization algorithms and utilities for training machine learning models. `Optax` was purely chosen as it's the default optimization library for `JAX` and there may be a better choice for this project, but it was not explored due to time constraints.

¹A pytree is a nested data structure in `JAX`: <https://docs.jax.dev/en/latest/pytrees.html>

3 Genel Theory

In this section, we will present the type of forward algorithm used to compute the likelihood and the pseudo residuals used to evaluate the model fit.

3.1 Forward Algorithm

The forward algorithm implemented is called recursive filtering (Møller et al., 2024) as it constructs the current statedistribution $p(X_t | Y_{1:t})$ recursively from the previous state distribution $p(X_t | Y_{1:t-1})$.

Algorithm 1 Recursive Filtering Forward Algorithm

Require: Observations $\mathbf{y} = (y_1, \dots, y_T)$, covariates $\mathbf{x} = (x_1, \dots, x_T)$, initial state distribution $\mathbf{u}_0 \in \mathbb{R}^{1 \times N}$, HMM parameters θ

Ensure: Predicted distributions $\{\mathbf{u}_t\}$, filtered distributions $\{\mathbf{u}_{t|t}\}$, step likelihoods $\{f_t\}$

```

1: for  $t = 1, \dots, T$  do
2:    $\Gamma_t \leftarrow \text{TRANSITIONMATRIX}(\theta, t, y_t, x_t)$  ▷  $N \times N$  transition matrix
3:    $\mathbf{u}_{t|t-1} \leftarrow \mathbf{u}_{t-1|t-1} \cdot \Gamma_t$  ▷ Predicted state distribution,  $1 \times N$ 
4:    $\mathbf{g}_t \leftarrow \text{DENSITY}(\theta, t, y_t, x_t)$  ▷ Emission densities for all states,  $1 \times N$ 
5:    $f_t \leftarrow \sum_{i=1}^N [\mathbf{u}_t \odot \mathbf{g}_t]_i$  ▷ Marginal likelihood at time  $t$ 
6:    $f_t \leftarrow \max(f_t, 10^{-10})$  ▷ Clip to avoid division by zero
7:    $\mathbf{u}_{t|t} \leftarrow \frac{\mathbf{u}_t \odot \mathbf{g}_t}{f_t}$  ▷ Filtered state distribution (normalized)
8: end for
9: return  $\{\mathbf{u}_t\}, \{\mathbf{u}_{t|t}\}, \{f_t\}$ 

```

where $\odot$ denotes element-wise multiplication of vectos.

The log-likelihood is obtained by summing the log of the step likelihoods:

$$\mathcal{L}(\theta) = \sum_{t=1}^T \log f_t \quad (1)$$

3.2 Forecast Pseudo Residuals

A method to access the goodness-of-fit of the HMM is to compute the pseudo residuals. For a introduction to pseudo residuals, see chapter 6.2 in (Zucchini et al., 2016).

The Forecast Pseudo Residuals used is the standard normal inverse of the cummulative distribution function (CDF) of the one-step-ahead predictive distribution. If the model is is good, the pseudo residuals should be approximately standard normally distributed.

Briefly desribed, a forecast pseudo residual for a HMM is defined as

$$z_{t+1} = \Phi^{-1}(F_Y(y_{t+1} | Y_{1:t})) \quad (2)$$

where Φ^{-1} is the inverse CDF of the standard normal distribution and F_Y is the predictive CDF of y_{t+1} given $Y_{1:t}$, defined as

$$F_Y(y_{t+1} | Y_{1:t}) = \sum_{i=1}^N u_{t+1|t}^i F_i(y_{t+1} | X_t = i) \quad (3)$$

where X_t is the hidden state.

Because z_{t+1} now should follow a standard normal distribution, we can use normality tests to assess the goodness-of-fit of the model.

In this project, we use the QQ-plot to visually assess the normality of the pseudo residuals.

3.3 Test Statistics

Other methods to assess the goodness of fit of the HMM model include the AIC (Wikipedia contributors, 2026a) and BIC (Wikipedia contributors, 2026b) test statistics. These are defined as

$$\text{AIC} = 2k - 2\mathcal{L} \quad (4)$$

$$\text{BIC} = k \log(n) - 2\mathcal{L} \quad (5)$$

where k is the number of parameters in the model, n is the number of observations and $\mathcal{L}$ is the log-likelihood of the model.

The AIC and BIC test statistics are used to compare different models, where a lower value initiates a better fit.

However, a lot of the models implemented are nested models, so a Likelihood Ratio Test (LRT) (Wikipedia contributors, 2026c) can also be used to compare the models. The LRT test statistic is defined as

$$\text{LRT} = 2(\mathcal{L}_1 - \mathcal{L}_0) \quad (6)$$

where $\mathcal{L}_1$ is the log-likelihood of the more complex model and $\mathcal{L}_0$ is the log-likelihood of the simpler model.

The LRT test statistic follows a chi-squared distribution with degrees of freedom equal to the difference in the number of parameters between the two models.

4 Models & Results

For general theory on HMM's, see (Zucchini et al., 2016) and (Møller et al., 2024) chapter 11. All the models implemented are 4 state Gaussian HMMs (emission density is a Gaussian distribution) and we have kept the the mean value of state 1 fixed at 400 as this is the baseline CO₂ level in the room when no one is present and all windows are closed. This is also shown in the figure 7.

Five models are considered, organised along two axes: the order of the AR component in the emission and the order of the latent Markov chain. Their relationship is shown in Figure 1.

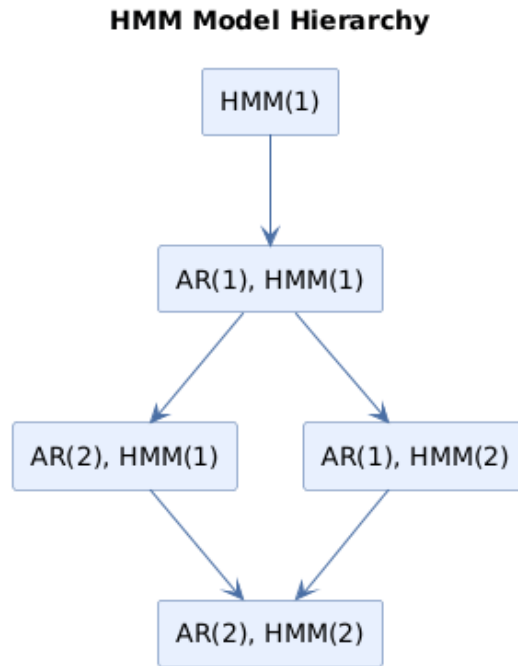


Figure 1: Hierarchy of the HMM models considered in this section. Each extension adds either an autoregressive emission component (AR order) or a higher-order latent chain (HMM order), culminating in the AR(2) second-order HMM.

For each level in the hierarchy tree in 1 starting from HMM(1), a method for capturing the temporal structure of the data is added. The HMM(1) is the ordinary HMM, which does not capture any temporal structure in the data.

4.1 Ordinary HMM model

4.1.1 Theory

The ordinary HMM model is the simplest model implemented in this project and the one most commonly used in the literature. The transition probabilities are constant and do not depend on time, the observations or any covariates, and so does the emission density.

For this model, we want to estimate the parameters $\theta = \{\Gamma, \mu, \sigma\}$ where $\Gamma \in \mathbb{R}^{n \times n}$ is the transition matrix, $\mu \in \mathbb{R}^n$ is the mean vector of the Gaussian emission density and $\sigma \in \mathbb{R}^n$ is the variance vector of the Gaussian emission density. Because the markov chain is finite, discrete and homogeneous, a unique strictly positive stationary distribution δ exists (chapter 1.3.2 (Zucchini et al., 2016)) and is given by

$$\delta = \mathbf{1}^T (I - \Gamma + E)^{-1} \quad (7)$$

where E is a matrix of ones.

To handle the natural bounds on the standard deviation parameters and transition probabilities, the parameters were reparameterized as follows:

$$\sigma_i = \exp(\eta_i) \quad (8)$$

For the transition matrix, the $n(n-1)$ off-diagonal entries are parametrized. The diagonal is used as the reference category with log-odds fixed to zero such that a transition matrix on the from

$$\xi = \begin{bmatrix} - & \xi_{12} & \cdots & \xi_{1n} \\ \xi_{21} & - & \cdots & \xi_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \xi_{n1} & \xi_{n2} & \cdots & - \end{bmatrix} \quad (9)$$

Concretely, the unconstrained parameters are stored as a matrix $\xi \in \mathbb{R}^{n \times (n-1)}$ that holds the off-diagonal logits. The transition matrix is then obtained by exponentiating these logits in the off-diagonal positions of an identity matrix (so the diagonal is $\exp(0) = 1$) and row-normalizing:

$$\Gamma_{ij} = \begin{cases} \frac{\exp(\xi_{ij})}{1 + \sum_{k \neq i} \exp(\xi_{ik})}, & j \neq i, \\ \frac{1}{1 + \sum_{k \neq i} \exp(\xi_{ik})}, & j = i. \end{cases} \quad (10)$$

The unconstrained parameters $\eta \in \mathbb{R}^n$ and $\xi \in \mathbb{R}^{n \times (n-1)}$ are the quantities that are optimized during the fitting process.

The μ parameters are not bounded, but to avoid label switching for the gaussian distributions, the mean parameters are ordered as $\mu_1 < \mu_2 < \mu_3 < \mu_4$ by the reparameterization

$$\begin{aligned} \mu_1 &= \zeta_1 \\ \mu_i &= \zeta_1 + \sum_{k=2}^i \exp(\zeta_k) \end{aligned} \quad (11)$$

where $\zeta \in \mathbb{R}^n$ is the unconstrained parameter vector that is optimized during the fitting process.

4.1.2 Results

The model was fitted on the CO₂ observations by minimising the negative log-likelihood with L-BFGS, with μ_1 frozen at 400 ppm as discussed above. The estimated transition matrix is shown in Table 1 and the estimated emission parameters in Table 2.

	State 1	State 2	State 3	State 4
State 1	0.8496	0.1476	0.0029	0.0000
State 2	0.0828	0.8486	0.0686	0.0000
State 3	0.0004	0.1605	0.6841	0.1550
State 4	0.0000	0.0000	0.0718	0.9282

Tabel 1: Estimated transition matrix for the ordinary 4-state HMM.

The transition matrix has a strongly dominant diagonal with probabilities between 0.68 and 0.93. Some transition is not possible in the matrix as to go from state 1 or 2 to state 4 or from state 4 to state 1 or 2. We see that for each state, there is a least 0.8 probability of switching state and this tells us that the model doesn't get stuck in 1 state for too long or forever.

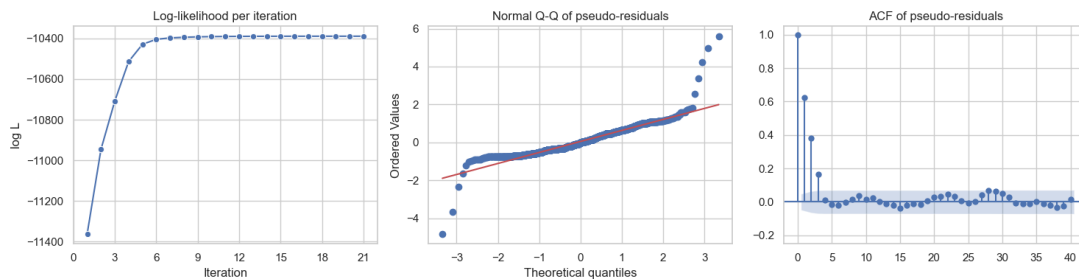
State	Mean ($\hat{\mu}$)	Std. Dev. ($\hat{\sigma}$)
State 1	400.000	21.748
State 2	535.684	76.573
State 3	889.928	110.981
State 4	1306.484	197.410

Tabel 2: Estimated emission parameters (Gaussian) for the ordinary 4-state HMM.

The four estimated state means in Table 2 (400, 536, 890, 1306 ppm) admit a natural interpretation as a *baseline*, *low*, *medium* and *high* CO₂ levels. The emission standard deviations grow with the mean, which makes sense as the we would expect more volatility in the CO₂ signal when people are present and closing and opening windows.

From these two tables, a interpretation of the states can be that

- state 1 is the baseline state where occupants has not entered the room in a long time.
- state 4 is the when the occupants has been in the room for a long time and the CO₂ level is very high.
- state 2 and 3 are intermediate states where the occupants has entered the room but not for a long time or windows has been opnend before reaching state 4, so the CO₂ level is higher than the baseline but not as high as in state 4.



Figur 2: Diagnostic plots for the ordinary 4-state HMM. Left: log-likelihood per iteration. Centre: normal Q-Q plot of pseudo-residuals. Right: ACF of pseudo-residuals.

Figure 2 summarises the fit. The log-likelihood trace (left panel) shows that the optimiser converges quickly towards the minimum of the negative log-likelihood. Almost all of the improvement happens in the first few iterations and the optimizer only keeps going due to relative tolerance in the negative log-likelihood is set to $\epsilon = 10^{-6}$. The Q-Q plot (centre) shows that the pseudo-residuals are approximately normal in the middle with some deviation in the tails. The ACF plot however clearly shows that there is autocorrelation in the pseudo-residuals as the 3 first lags are decaying slowly. This indicates that we need to introduce temporal structure between the observations to capture the dynamics of the data better. This motivates the introduction of the AR-HMM in the next section, which adds an autoregressive component to the emission density.

4.2 AR(1)-HMM

4.2.1 Theory

The Autoregressive HMM (AR(1)-HMM) extends the ordinary HMM by letting the emission distribution depend on the previous observation. Conditional on being in state i at time t , the observation is modelled as

$$Y_t \mid X_t = i, Y_{t-1} = y_{t-1} \sim \mathcal{N}(\mu_i + \phi_i(y_{t-1} - \mu_i), \sigma_i^2), \quad (12)$$

Therefore, each states has its own AR(1) coefficient $\phi_i \in (-1, 1)$ in addition to a mean and variance. This may be enough to capture the temporal correlation of the CO₂ signal that the ordinary HMM left in the pseudo-residuals. The transition matrix Γ is parametrised exactly as in the ordinary HMM, and the AR coefficients are reparametrised as $\phi_i = 2\sigma(\psi_i) - 1$, where $\sigma(z) = (1 + e^{-z})^{-1}$ is the logistic sigmoid, to enforce the stationarity constraint $|\phi_i| < 1$.

The unconstrained parameter $\psi_i \in \mathbb{R}$ is the one passed to the optimiser, with inverse mapping $\psi_i = \text{logit}\left(\frac{\phi_i + 1}{2}\right)$.

4.2.2 Results

The ordinary-HMM estimates were used as a warm start for Γ , μ and σ , and the AR coefficients were initialised at $\phi_i = 0.2$ for all states. The estimated transition matrix is shown in Table 3 and the estimated emission parameters in Table 4.

	State 1	State 2	State 3	State 4
State 1	0.7521	0.1685	0.0016	0.0779
State 2	0.2698	0.5960	0.0633	0.0709
State 3	0.0516	0.1027	0.8457	0.0000
State 4	0.0067	0.0492	0.0153	0.9289

Tabel 3: Estimated transition matrix for the AR-HMM.

The transition matrix strongly dominant in the diagonal, with probabilities between 0.74 and 0.97. The off-diagonal pattern is similar to the ordinary HMM, but now only 1 transition is impossible, which is state 3 to state 4. We switched self transition probabilities between state 2 and 3, meaning AR(1)-HMM says it's more likely to stay in state 2 than state 3, while the ordinary HMM says the opposite.

The estimated state means in Table 4 are similar to the ordinary HMM, with a clear ordering from low to high. We see that state 4 mean is close the HMM state 4 mean, but state 2 and 3 means are closer together than in the ordinary HMM. This validates the interpretation of these two states being transitional states between the baseline and the high-CO₂ state.

State	Mean ($\hat{\mu}$)	Std. Dev. ($\hat{\sigma}$)	AR coeff. ($\hat{\phi}$)
State 1	400.000	12.501	0.6235
State 2	509.326	77.704	0.3659
State 3	560.285	11.006	0.9462
State 4	1366.661	102.836	0.8318

Tabel 4: Estimated emission parameters for the AR-HMM (Gaussian with AR(1) residuals).

All four AR coefficients are positive and substantial, ranging from $\hat{\phi}_2 = 0.37$ up to $\hat{\phi}_3 = 0.95$; the high-CO₂ state (state 4) also has a strong AR component ($\hat{\phi}_4 = 0.83$). This confirms that a non-trivial amount of the temporal structure in the observations is captured by the AR(1) component of the emission density. The positive AR coefficients also make sense as we would expect the CO₂ signal at a timestep t to be positively correlated with the CO₂ ppm at time $t - 1$ by logical thinking.

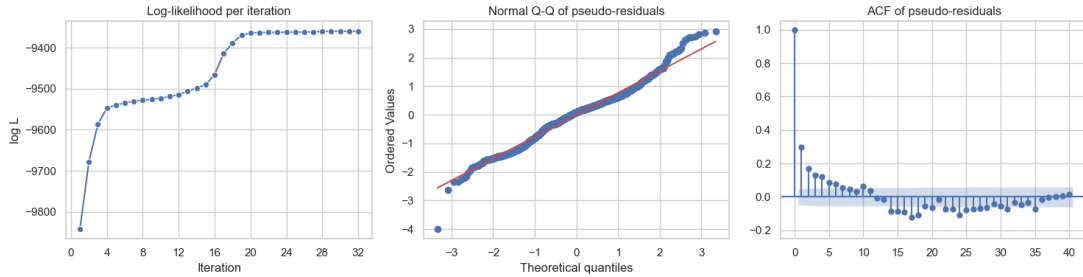


Figure 3: Diagnostic plots for the AR-HMM. Left: log-likelihood per iteration. Centre: normal Q-Q plot of pseudo-residuals. Right: ACF of pseudo-residuals.

Figure 3 shows the diagnostics for the AR-HMM. The log-likelihood trace (left panel) shows that the optimiser converges after a few more iterations than for the ordinary HMM, which is expected as the AR-HMM has more parameters. The Q-Q plot of the pseudo-residuals (centre) shows a clear improvement in the normality of the residuals compared to the ordinary HMM, with only a single outlier in the lower tail and few more in the upper tail. The ACF however seems to have been worsened as there are more lags with significant residual spikes, and the first lags drops quickly to around 0.3 and then decays slowly. We also see that after lag 15, almost all of the lags are negative. This indicates that while the AR(1) component captures a lot of the temporal structure in the data, there is still some autocorrelation left in the residuals. This motivates the introduction of the AR(2)-HMM in the next section, which adds a second-order autoregressive component to the emission density.

4.3 AR(2)-HMM

4.3.1 Theory

The Autoregressive of order 2 HMM (AR(2)-HMM) extends the AR-HMM by letting the autoregressive component depend on the two most recent observations rather than just one. Conditional on being in state i at time t , the observation is modelled as

$$Y_t \mid X_t = i, y_{t-1}, y_{t-2} \sim \mathcal{N}(\mu_i + \phi_{i,1}(y_{t-1} - \mu_i) + \phi_{i,2}(y_{t-2} - \mu_i), \sigma_i^2), \quad (13)$$

so each state now has a pair of AR coefficients $(\phi_{i,1}, \phi_{i,2})$ in addition to a mean and variance.

4.3.2 Results

The AR-HMM estimates were used as a warm start, and the second-order AR coefficients were initialised at $\phi_{i,2} = 0.2$ for all states. The estimated transition matrix is shown in Table 5 and the estimated emission parameters in Table 6.

	State 1	State 2	State 3	State 4
State 1	0.7115	0.2065	0.0000	0.0820
State 2	0.2848	0.5479	0.1055	0.0618
State 3	0.0216	0.1325	0.8459	0.0000
State 4	0.0061	0.0453	0.0171	0.9315

Table 5: Estimated transition matrix for the AR(2)-HMM.

The transition matrix is very close to the one obtained for the AR-HMM. Adding a second lag therefore does not noticeably change the between-state dynamics of the model.

State	Mean ($\hat{\mu}$)	Std. Dev. ($\hat{\sigma}$)	AR coeff. ($\hat{\phi}_1$)	AR coeff. ($\hat{\phi}_2$)	$\hat{\phi}_1 + \hat{\phi}_2$
State 1	400.000	10.765	0.7027	-0.0781	0.6247
State 2	525.630	72.083	0.5941	-0.2109	0.3832
State 3	716.015	16.158	0.9596	0.0565	1.0161
State 4	1299.660	97.603	0.9997	-0.2093	0.7904

Table 6: Estimated emission parameters for the AR(2)-HMM (Gaussian with AR(2) residuals).

The first-order AR coefficients in Table 6 remain strongly positive across all states (0.59–1.00), while the second-order coefficients are small and mostly negative, ranging from -0.21 to 0.06 . The sum of the AR coefficients resemble the AR(1) coefficients of the previous model, but the first lag is much higher and the second lag is negative, with except for state 3. This tells us that the observation at time t is more positively correlated with the observation at time $t-1$ and that the observation at time $t-2$ has a small negative effect on the observation at time t .

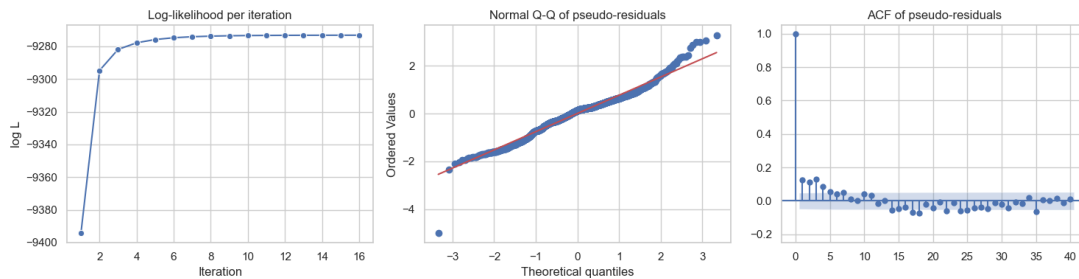


Figure 4: Diagnostic plots for the AR(2)-HMM. Left: log-likelihood per iteration. Centre: normal Q-Q plot of pseudo-residuals. Right: ACF of pseudo-residuals.

Figure 4 shows the diagnostics for the AR(2)-HMM. The log-likelihood converges more quickly to its optimum than AR(1)-HMM, even though we have introduced an additional parameter, and the Q-Q plot of the pseudo-residuals is similar to that of the AR(1)-HMM. The ACF plot however shows a clear improvement in the autocorrelation of the residuals, as fewer lags are outside the confidence band and the first lags are smaller than in the AR(1)-HMM. However, the pattern of the ACF is similar to the AR(1)-HMM and there seem to be a pattern we haven't captured yet.

This motivates the introduction of the second-order AR(1)-HMM(2) in the next section, which makes introduces a second order markoc chain.

4.4 AR(1)-HMM(2)

4.4.1 Theory

The Autoregressive second-order HMM (AR(1)-HMM(2)) reformulates the Markov assumption on the latent chain. Now, the next state depends on the two most recent latent states rather than just one. The transition probabilities are therefore indexed by a pair of states,

$$\Gamma_{(i,j),k} = P(X_{t+1} = k \mid X_{t-1} = i, X_t = j), \quad (14)$$

which gives an $n^2 \times n$ tensor of transition probabilities (still row-normalised in k). The emission density is again Gaussian, with the mean depending only on the current state $\mu(x_t)$, but the standard deviation allowed to depend on both the previous and the current state $\sigma(x_{t-1}, x_t)$. When implementing the second-order chain, the state space is augmented to pairs $S_k = (x_{i,t-1}, x_{j,t})$ where $x_{t-k,i}$ is the state i at time $t-k$. $S \in \mathbb{R}^M$ is recognized a the state distribuion. We can the define the possible transitisinos in the augmented transition matrix $\Gamma^* \in \mathbb{R}^{M \times M}$ as

$$\Gamma_{(i,j),(j,k)}^* = \Gamma_{(i,j),k}^* \quad (15)$$

where k is the next state given the current state (i, j) . This is implmented in the code by checking the that the current state j is the previous state in the next state. We only store the $n^2 \times n$ log transition probability tensor ξ and not the full $M \times M$ transition matrix Γ^* . When constructing the full transition matrix Γ^* , we fill the entries according to 15 and set the rest of the entries to a highly negative value. We then apply the row-wise softmax to get a valid transition matrix.

4.4.2 Results

The estimated transition tensor is shown in Table 7 and the estimated emission parameters in Table ??.

The second-order transition table shows that for most state state pairs (i, j) , want to transition

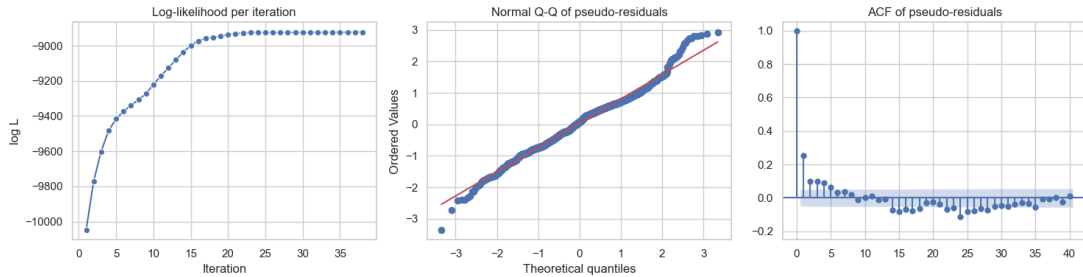


Figure 5: Diagnostic plots for the second-order HMM. Left: log-likelihood per iteration. Centre: normal Q-Q plot of pseudo-residuals. Right: ACF of pseudo-residuals.

Figure 5 shows the diagnostics for the second-order HMM. The log-likelihood converges quickly toward the minimum of the negative log-likelihood, in the same way as for the previous models. The Q-Q plot of the pseudo-residuals is comparable to the ordinary HMM, and the ACF of the residuals still shows a slow decay with several lags well outside the confidence band. The second-order extension therefore improves the flexibility of the latent dynamics, but — because it does not introduce any within-state autocorrelation in the observations — it does not remove the residual time-series structure in the same way the AR-HMM does.

From (s_{t-1}, s_t)	$s_{t+1} = 0$	$s_{t+1} = 1$	$s_{t+1} = 2$	$s_{t+1} = 3$
(0, 0)	0.7633	0.1270	0.0208	0.0889
(0, 1)	0.0705	0.9294	—	—
(0, 2)	0.7629	—	0.0001	0.2370
(0, 3)	0.7065	0.0663	0.1706	0.0565
(1, 0)	0.6903	0.0497	—	0.2600
(1, 1)	0.1675	0.6206	0.0153	0.1966
(1, 2)	—	0.0738	—	0.9262
(1, 3)	0.0242	0.1217	—	0.8542
(2, 0)	0.1370	—	0.8630	—
(2, 1)	0.5383	0.3560	0.1056	0.0001
(2, 2)	0.0425	—	0.0010	0.9564
(2, 3)	0.5295	0.1390	0.3313	0.0001
(3, 0)	0.1148	—	0.3133	0.5719
(3, 1)	0.0124	0.0767	0.3351	0.5757
(3, 2)	—	0.9998	—	0.0002
(3, 3)	—	0.1688	0.0135	0.8177

Tabel 7: Estimated transition probabilities for the AR(1) second-order HMM. Each row is a state pair (s_{t-1}, s_t) ; columns give the next-state probability $P(s_{t+1} | s_{t-1}, s_t)$. A dash (—) indicates zero or negligible probability.

4.5 AR(2) second-order HMM

4.5.1 Theory

The AR(2) second-order HMM combines the two previous extensions: the latent chain is second-order (transitions depend on the pair (x_{t-1}, x_t)) and the emission density carries an AR(2) component whose parameters also depend on the pair (x_{t-1}, x_t) . Conditional on $(X_{t-1}, X_t) = (i, j)$ and the two most recent observations, the observation density is

$$Y_t | (i, j), y_{t-1}, y_{t-2} \sim \mathcal{N}(\mu_j + \phi_{(i,j),1}(y_{t-1} - \mu_j) + \phi_{(i,j),2}(y_{t-2} - \mu_j), \sigma_{(i,j)}^2). \quad (16)$$

The mean depends only on the current state x_t , while $\sigma_{(i,j)}^2$ and the AR coefficients $(\phi_{(i,j),1}, \phi_{(i,j),2})$ are pair-specific. The transition tensor is the same as in the second-order HMM, and the same reparametrisations are used for the transition rows, the log-variances and the AR coefficients. The full parameter vector to be estimated is $\theta = \{\Gamma_{(\cdot,\cdot),\cdot}, \mu, \sigma^2, \phi_1, \phi_2\}$.

4.5.2 Results

The AR(2) second-order HMM was fitted on the same CO₂ observations, with μ_1 frozen at 400 ppm and the previous models used as a warm start. The estimated transition tensor is shown in Table 8 and the estimated emission parameters in Table 9.

The transition tensor follows the same overall pattern as the second-order HMM: rows like (1, 1), (2, 2) and (3, 3) keep most of their mass on the current state, while several rare two-step pairs end up with all the mass on a single successor. The presence of an AR(2) emission shifts some mass off the diagonal compared to the second-order HMM (e.g. (0, 0) now has $P(x_{t+1} = 2 | 0, 0) = 0.29$ versus essentially zero before), because the within-state autocorrelation already explains part of the persistence and the latent chain no longer needs as strong self-loops.

Table 9 reports the pair-specific emission parameters. The four state means again order the regimes (400, 489, 710, 1399 ppm). The first-order AR coefficients $\hat{\phi}_1$ are uniformly large and

From (s_{t-1}, s_t)	$s_{t+1} = 0$	$s_{t+1} = 1$	$s_{t+1} = 2$	$s_{t+1} = 3$
(0, 0)	0.7192	0.2242	—	0.0565
(0, 1)	0.2137	0.4312	0.3548	0.0002
(0, 2)	0.7218	—	—	0.2782
(0, 3)	0.8343	0.1020	—	0.0637
(1, 0)	0.6948	0.3049	—	0.0003
(1, 1)	0.1289	0.6793	0.0328	0.1590
(1, 2)	—	0.5243	—	0.4757
(1, 3)	—	0.1968	—	0.8032
(2, 0)	0.1582	—	0.8418	—
(2, 1)	0.3295	0.5295	0.1410	—
(2, 2)	0.0008	—	0.3577	0.6415
(2, 3)	0.0409	0.1158	0.8432	0.0001
(3, 0)	0.0620	—	0.3651	0.5729
(3, 1)	—	0.0154	0.3636	0.6210
(3, 2)	0.1742	0.2723	—	0.5534
(3, 3)	—	0.1599	0.0272	0.8129

Tabel 8: Estimated transition probabilities for the AR(2) second-order HMM. Each row is a state pair (s_{t-1}, s_t) ; columns give the next-state probability $P(s_{t+1} | s_{t-1}, s_t)$. A dash (—) indicates zero or negligible probability.

positive (most are between 0.6 and 1.0), confirming that within-regime autocorrelation dominates the short-term dynamics regardless of x_{t-1} . The second-order coefficients $\hat{\phi}_2$ are mostly small and negative, in line with what was observed for the simpler AR(2)-HMM. The last column reports $\hat{\phi}_1 + \hat{\phi}_2$, which approximates the long-run persistence of the AR(2) component: a few pairs — notably (1, 3) with $\hat{\phi}_1 + \hat{\phi}_2 = 1.23$ — fall outside the stationarity region. These non-stationary pair-specific AR processes sit on transitions that are sparsely populated in the data and should be interpreted as weakly identified artifacts of the fit, in the same way as the inflated σ values in the second-order HMM.

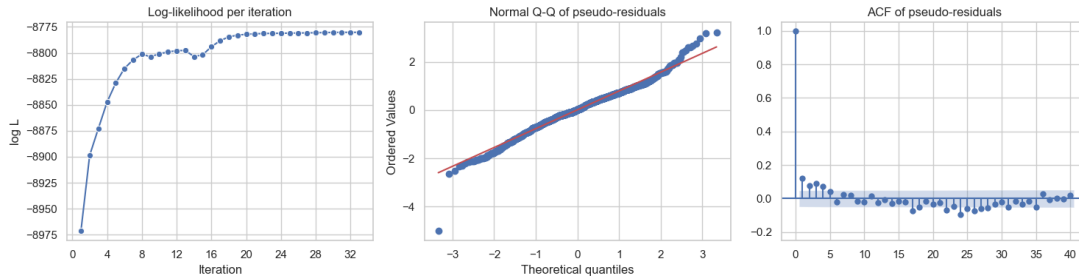


Figure 6: Diagnostic plots for the AR(2) second-order HMM. Left: log-likelihood per iteration. Centre: normal Q-Q plot of pseudo-residuals. Right: ACF of pseudo-residuals.

Figure 6 shows the diagnostics for the AR(2) second-order HMM. The log-likelihood again converges quickly to its optimum. The Q-Q plot of the pseudo-residuals is close to the diagonal, and the ACF is essentially white noise: combining a second-order chain with an AR(2) emission gives the best residual diagnostics of the four models, even though the marginal improvement over the plain AR-HMM is small.

s_{t-1}	s_t	$\hat{\mu}$	$\hat{\sigma}$	$\hat{\phi}_1$	$\hat{\phi}_2$	$\hat{\phi}_1 + \hat{\phi}_2$
0	0	400.000	7.818	0.7825	-0.0960	0.6865
1	0	400.000	12.809	0.4320	-0.1232	0.3088
2	0	400.000	6.407	1.0000	-0.0030	0.9970
3	0	400.000	126.469	1.0000	0.0524	1.0524
0	1	437.521	38.283	-0.2797	0.1697	-0.1100
1	1	437.521	16.359	0.5665	-0.0767	0.4897
2	1	437.521	67.879	0.7221	-0.1645	0.5576
3	1	437.521	113.350	1.0000	-0.1702	0.8298
0	2	570.631	6.012	0.9285	-0.0006	0.9279
1	2	570.631	98.402	-0.1690	0.4549	0.2859
2	2	570.631	0.000	0.9594	0.0327	0.9922
3	2	570.631	31.589	0.9997	0.0469	1.0466
0	3	1436.337	174.176	1.0000	-0.1229	0.8771
1	3	1436.337	94.026	0.9204	-0.0704	0.8500
2	3	1436.337	40.134	0.9998	0.0017	1.0015
3	3	1436.337	56.351	0.9458	-0.1708	0.7750

Tabel 9: Estimated emission parameters for the AR(2) second-order HMM. $\hat{\sigma}$ and the AR(2) coefficients $\hat{\phi}_1, \hat{\phi}_2$ depend on (s_{t-1}, s_t) . A sum $\hat{\phi}_1 + \hat{\phi}_2 > 1$ indicates a non-stationary AR process for that state pair.

5 Discussion

To compare the five fitted models we use the likelihood-ratio test (LRT) together with AIC and BIC. The LRT applies to nested pairs of models: for a base model with log-likelihood ℓ_0 and an expanded model with log-likelihood ℓ_1 the statistic

$$\Lambda = 2(\ell_1 - \ell_0) \quad (17)$$

is asymptotically χ^2 -distributed with degrees of freedom equal to the difference in the number of free parameters, under the null that the base model is adequate. Small p -values therefore favour the expanded model. AIC and BIC do not require nesting; lower values indicate a better likelihood-complexity trade-off, with BIC penalising parameters more heavily than AIC.

Table 10 reports the per-model statistics for all five models in the hierarchy of Figure 1. Table 11 reports the LRT, p -value, Δ AIC and Δ BIC for each edge of the hierarchy, where the difference is taken as expanded minus base so that negative values favour the expanded model.

Model	#Params	LogLik	AIC	BIC
HMM(1)	20	-10388.87	20817.73	20926.10
AR(1), HMM(1)	24	-9359.73	18767.45	18897.47
AR(2), HMM(1)	28	-9273.19	18602.38	18754.06
AR(1), HMM(2)	84	-9130.90	18429.80	18884.88
AR(2), HMM(2)	100	-9045.40	18290.79	18832.49

Tabel 10: Per-model log-likelihood, AIC and BIC for the five fitted HMMs.

Base Model	Expanded Model	LRT	df	<i>p</i> -value	Δ AIC	Δ BIC
HMM(1)	AR(1), HMM(1)	2058.28	4	0.0000	-2050.28	-2028.61
AR(1), HMM(1)	AR(2), HMM(1)	173.07	4	0.0000	-165.07	-143.40
AR(1), HMM(1)	AR(1), HMM(2)	457.65	60	0.0000	-337.65	-12.60
AR(2), HMM(1)	AR(2), HMM(2)	455.59	72	0.0000	-311.59	78.43
AR(1), HMM(2)	AR(2), HMM(2)	171.01	16	0.0000	-139.01	-52.34

Tabel 11: Likelihood-ratio tests for each nested pair from the model hierarchy (Figure 1). Negative Δ AIC/ Δ BIC favour the expanded model.

6 Conclusion

This project investigated whether room occupancy can be inferred from a single CO₂ sensor by modelling the observed series with a Hidden Markov Model. Five nested 4-state Gaussian HMMs were fitted, organised along the hierarchy of Figure 1: a baseline HMM(1), an AR(1) and AR(2) emission extension, a second-order latent chain with AR(1) emissions, and the fully combined AR(2), HMM(2) model. All models were implemented and fitted using `JAX`, `EQUINOX` and `Optax`.

The likelihood-ratio tests in Table 11 reject every base model in favour of its expanded counterpart at $p < 10^{-4}$, confirming that both adding autoregressive structure to the emissions and relaxing the first-order Markov assumption produce statistically significant gains in fit. The single largest improvement is the introduction of an AR(1) emission on top of the ordinary HMM, with a likelihood-ratio statistic of $\Lambda \approx 2058$ on four degrees of freedom and a drop in AIC of more than two thousand units. This indicates that most of the temporal structure left unexplained by the baseline HMM is autoregressive in nature, not a deficiency of the first-order latent chain.

AIC and BIC, however, disagree on the recommended model. Table 10 shows that AIC decreases monotonically along the hierarchy and is minimised by AR(2), HMM(2), reflecting the genuine gain in log-likelihood from the largest model. BIC, which penalises additional parameters more heavily, instead selects AR(2), HMM(1): both second-order models add many parameters (one variance and AR coefficient per ordered state pair) without enough likelihood improvement to offset the penalty on this sample size. The takeaway is that the second-order chain pays for itself in fit but not in parsimony.

For practical use as an occupancy model, where interpretability and a manageable number of per-state parameters are valuable, AR(2), HMM(1) is the recommended model: it captures the within-regime autocorrelation that the plain HMM cannot, while keeping the latent dynamics first-order and the number of free parameters modest. AR(2), HMM(2) remains the strict best-fit model and is the appropriate choice when raw likelihood is the only criterion. Possible extensions to either model — covariate-dependent transitions, richer emission families, or further parameter tying in the second-order chain — are discussed in the Future Work section.

7 Future Work

Litteratur

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8 Appendix

8.1 Source Code Repository

The repository is hosted on GitHub and the source code can be found there with the data as well: <https://github.com/Maccen26/hmm-project>

8.2 Source code & Envoriment

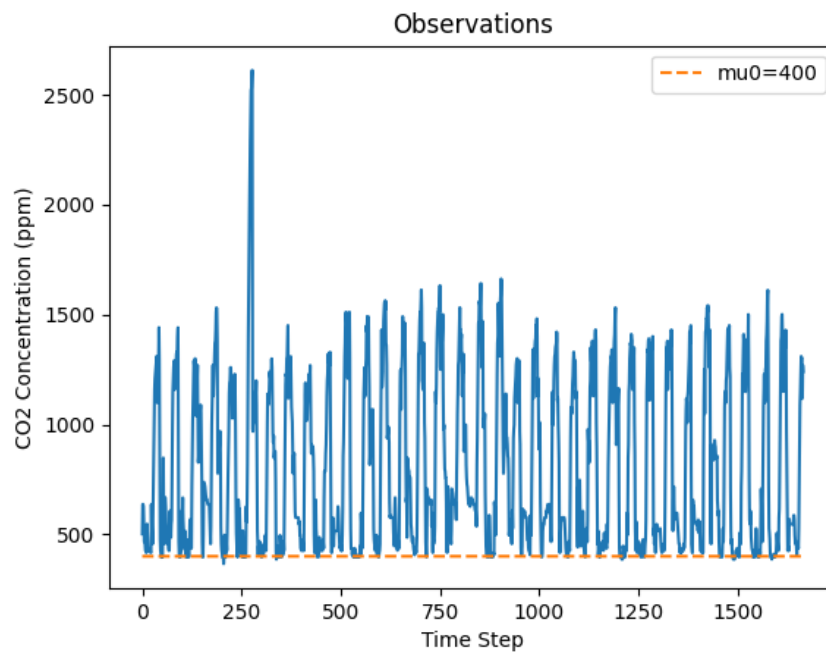
This code was written with Python (version 3.11) and uv (version 0.9.7) (Astral Software Inc., 2024) was used as a package manager. The libraries used are listed in Table 12.

Library	Version
dotenv	$\geq 0.9.9$
equinox	$\geq 0.13.7$
jax	$\geq 0.9.0.1$
matplotlib	$\geq 3.10.8$
numpy	$\geq 1.23.5, < 2.3$
optax	$\geq 0.2.8$
pandas	$\geq 3.0.0$
seaborn	$\geq 0.13.2$
statsmodels	$\geq 0.14.6$

Tabel 12: Python libraries used in this project.

The source code is organized in a way such that the directory **src** contains the source code for the project, the directory **data** contains the data used for training and testing the model, the directory **tests** contains tests for the source code and the directory **drivers** contains scripts for running the model and generating the results used in the report.

8.3 CO₂ Data



Figur 7: Observed CO₂ concentration (ppm) over time.

8.4 Class Diagrams

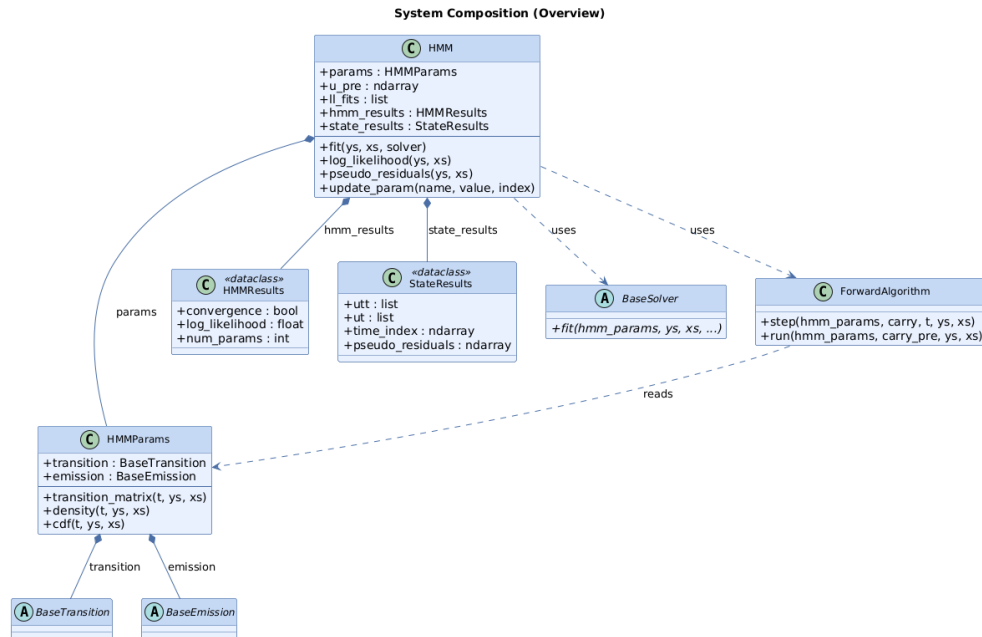


Figure 8: System composition overview. The user-facing **HMM** class owns an **HMMParams** instance composed of a **BaseTransition** and a **BaseEmission**. Fitting is delegated to a **BaseSolver** subclass and state inference is handled by **ForwardAlgorithm**, producing **HMMResults** and **StateResults**.

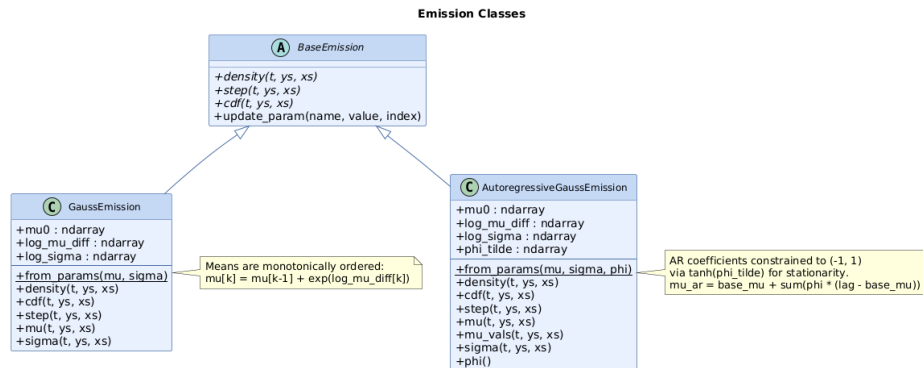


Figure 9: Emission class hierarchy. **GaussEmission** models state-dependent Gaussian distributions with monotonically ordered means enforced via cumulative exponentiation of **log_mu_diff**. **AutoregressiveGaussEmission** extends this with $AR(p)$ dynamics, where coefficients are stored unconstrained (**phi_tilde**) and mapped to $(-1, 1)$ via $\tanh(\phi_tilde)$ to guarantee stationarity.

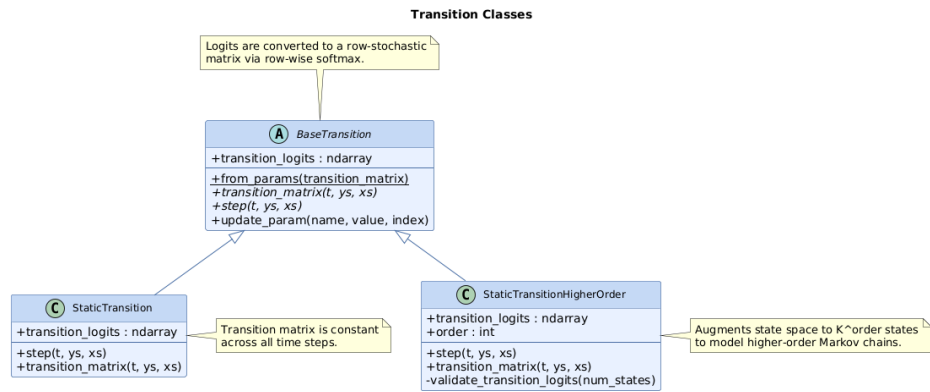


Figure 10: Transition class hierarchy. Both implementations store parameters as logits and convert them to a row-stochastic matrix via row-wise softmax. **StaticTransition** uses a constant $K \times K$ matrix, while **StaticTransitionHigherOrder** augments the state space to K^{order} states to represent a higher-order Markov chain.

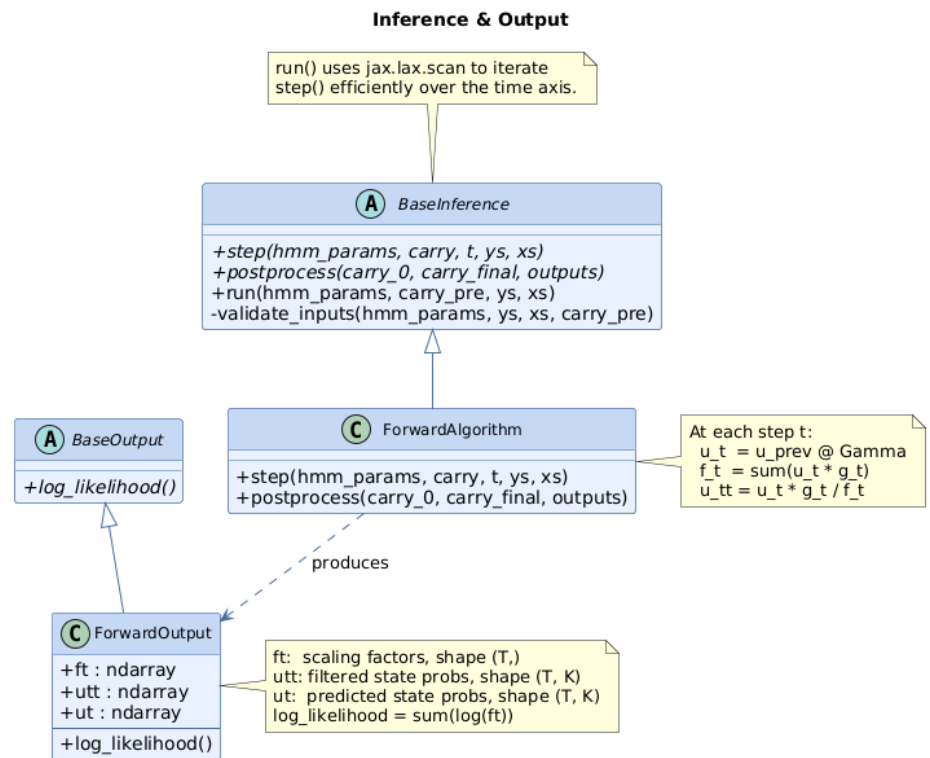


Figure 11: Inference and output class hierarchy. **BaseInference.run()** iterates `step()` across the time axis using `jax.lax.scan`. **ForwardAlgorithm** implements the forward filtering pass and produces a **ForwardOutput** storing the scaling factors f_t , filtered state probabilities $u_{t|t}$, and predicted state probabilities u_t . The log-likelihood is computed as $\sum_t \log f_t$.

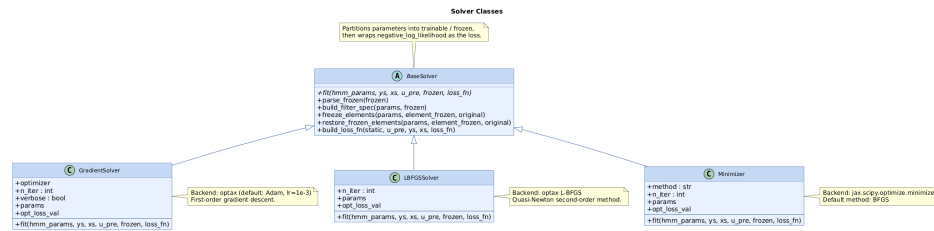


Figure 12: Solver class hierarchy. **BaseSolver** partitions parameters into trainable and frozen subsets and wraps the negative log-likelihood as the loss function. The three concrete solvers share the same interface but differ in their optimization backend: **GradientSolver** uses first-order gradient descent via **optax**, **LBFGSSolver** uses quasi-Newton L-BFGS via **optax**, and **Minimizer** wraps **jax.scipy.optimize.minimize**.

8.5 Class Design

The **HMM** class (8) is the main entry point of the system. Its responsibility is to act as a wrapper for the other components and to provide a simple interface for training and inference. It is dependent on **Emissions**, **Transitions**, **HMMParams**, and **Solvers**, which are composed together to form a complete HMM.

The **Emissions** (9) and **Transitions** (10) classes are abstract interfaces whose concrete implementations must be provided at runtime. This design choice is motivated by the structure of HMM variants: the differences between models in this project lie in how the density of an observation given a state is computed (emissions) and how the transition probabilities between states are computed (transitions). By separating these concerns into abstract interfaces, different HMM models can be composed without changing the rest of the system.

The **HMMParams** class acts as a container for the model parameters and as a wrapper for the methods implemented in the **Emissions** and **Transitions** classes. This is required by how **JAX** and **Equinox** work: parameters must be stored as pytrees to support automatic differentiation.

The **Inference** class (11) is an abstract interface responsible for implementing inference algorithms such as the forward algorithm, the backward algorithm, and the Viterbi algorithm. In this project, only the forward algorithm is implemented, but the design allows easy extension to other inference algorithms without modifying existing code.

The **Solvers** class (12) implements the optimization algorithms and depends on a loss function that takes the **HMMParams** and **Inference** classes as arguments. This dependency injection of the loss function decouples the optimizer from the objective, making it straightforward to change the loss function without altering the optimization algorithm.